Project Description

The **Design Document (Phase 1) must be submitted by February 23rd at 23:59**

The Implementation Report (Phase 2) must be submitted by Friday, April 2nd at 23:59

ONE and ONLY ONE Submission per Team through course Moodle.

Incomplete and inappropriate submissions will be penalized

Project Demos will begin on April 7th

RENTRUCK Inc.

RENTRUCK Company is a young Montreal firm, specialized in the rental of trucks with driver. It has a fleet of a hundred trucks of all types, and currently has a thousand regular customers, mostly businesses. Its business is growing strongly, so the CEO has decided to automate gradually the management of the company.

Description of the different activities related to the rental of trucks.

A. MISSIONS

The rentals have a variable duration. The drivers return necessarily to the garage with their truck at the week end. Each mission cannot exceed 5 days (Monday morning to Friday evening). So a rental is treated as N missions for a maximum duration of five days each.

A mission begins when the truck leaves the garage (even if the appointment is set at another location) and ends when the truck returns to the garage.

The number of missions generally varies from 100 to 800 per week.

The driver goes with the rented truck to the appointment fixed by the customer, and is available to him for the whole duration of the mission (until Friday evening at the latest).

A driver can drive any vehicle if s/he has the corresponding driving license: tourism, heavyweight, super heavyweight.

B. RESERVATIONS

Reservations are made by phone with the sales department. A customer can book a truck several weeks in advance for a period of maximum one year. The customer may of course modify or cancel all or part of a reservation, if it is requested no later than one week before starting a mission.

The form used by RENTRUCK for reservations is designed to decompose easily a reservation into missions. Thus, the information provided by the client can establish for each mission a record including:

- Number, name or business name and address of the client;
- Type of vehicle desired;
- The place of rendezvous;
- The date and time of appointment;
- The expected duration for vehicle and driver disposal.

The portfolio of missions may include up to 10,000 missions to realize.

C. PREPARATION OF MISSIONS

The staff working at the sales department then performs the following tasks:

- From the information provided by the client when booking, the staff determines the date and the start and end time of the missions;
- It assigns the drivers and vehicles for each mission. This assignment is currently done manually, using schedules wall. The manual process, though heavy, will be kept for some time: the atomization of the process will be the subject of further study.

This allows it to harmonize booking requests and opportunities for the company's service, for each mission.

They plan to computerize the subsequent procedure.

Thus the "mission sheets" definitely established will be daily filled out in the terminals (the same garage) in order to create for each mission a record in the database "**RENTRUCK**". They seek to minimize the volume of the seizure.

During the weekends, they plan to publish mission sheets for drivers. These documents define the missions that will be performed during the following week, due to a mission per sheet. They will also be used for billing of rentals.

We will not deal, in this context, the case of bookings made in midweek for the current week (midweek bookings are not included).

D. BILLING

A weekly invoice is sent to customers.

A customer can rent several vehicles simultaneously or successively. In this case, the invoice includes a line by mission.

Invoices are partly compiled from the mission sheet completed by drivers. They enroll in effect:

- Actual dates and hours of start and end of the mission, if they differ from what was expected;
- The value of the vehicle odometer in the garage before and after return.

The data necessary for billing on the “mission sheet” are daily entered from terminals. We look again to minimize the volume of the seizure.

The rental cost charged includes:

- A proportional fraction to the duration of the mission;
- And a proportional part to the browsed kilometers.

These two costs obviously depend on the rented vehicle.

The invoice payment is being made by credit card, cash or check.

MARKING GUIDELINES OF THIS PROJECT

IMPORTANT: **For this project, you MUST use the MySQL database of Concordia to execute queries and to interact from the system Web-based front-end or desktop system. This desktop must be transferred to the Concordia Server for testing it.**

DISTRIBUTION OF MARKS

Activities	Marks
Conceptual Model	13 pts.
Logical Model	10 pts.
Relational Model	10 pts.
Normalization of Relation (3NF) into logical model	10 pts.
Constraints	10 pts.
SQL Script of the creation of tables	10 pts.
Insertion of data (10 rows minimum for each **STRONG** table and 5 rows minimum for each **WEAK** table)	10 pts.
Implementation of the list of 11 Queries given below by providing the implementation details of your database system using the MySQL DBMS with a suitable user interfaces. #a : 2 pts. #b : 2 pts. #c : 3 pts. #d : 3 pts. #e : 4 pts. #f : 2 pts. #g : 2 pts. #h : 3 pts. #i : 2 pts. #j : 2 pts. #k : 2 pts.	27 pts.
TOTAL	**100%**

GUIDELINE FOR PREPARING PROJECT REPORT

Your project report **MUST**:

- Be properly formatted;
- Have group name, official names of the team members, and student ID's clearly printed on the cover;

Inappropriate submission will be marked zero (0).

At the end of the term, the teams will present their projects. You need to submit a technical document, called **Project_GroupId**, which describes your analysis and design. This document must be submitted electronically with the following information about the team members: the names and student IDs at ***https://fis.encs.concordia.ca/eas*****.**

Each team project. Each member is responsible for the entire project and at least a well-defined portion of the project, to be agreed by the team members.

Each team will have **15 minutes** to demonstrate their project. Every member of a team must be present during their project demo.

The document report **MUST** contain the following sections:

1. Conceptual Diagram.
2. Logical Diagram of your conceptual diagram provided in the section 1.
3. Relational diagram of your logical diagram provided in the section 2
4. Explain in plain English the normalization applied to the logical diagram.
5. Explain in plain English the different constraints used to the attributes of tables (NOT NULL, CHECK, primary key, foreign key, UNIQUE, etc.)
6. SQL Script of the creation of tables
7. Insertion of data for each table (10 rows minimum for each **STRONG** table and 5 minimum for each **WEAK** table)
8. Queries (Simples with different tables and Complex with inner SQL) that have been implemented to satisfy project requirements as follows :

   a. List of customers that are businesses (Enterprises or Companies).
   b. List of reservations whose reservation number is greater than 1.
   c. List of drivers and vehicles having participated in at least one mission.
   d. List of missions between March 11, 2026 and March 18, 2026 as well as the drivers and vehicles participating in these missions.
   e. The list of customers who have not paid their invoices.
   f. List of drivers who have driven 'GMC' brand vehicles.
   g. Which customers have invoices greater than $1000?
   h. List of customers with their number of associated invoices.
   i. What are the last names and first names of the drivers who have a mission between the following dates: February 1, 2026 and March 31, 2026 whose mileage (number of kilometers traveled) is more than 7000 km?

j. Write a transaction to update details of a mission, given the mission ID. (We expect the end date to change according).
k. Write a transaction to cancel a mission or part of a mission.

GUIDELINES FOR DEMO

All the members of a team **MUST** actively participate in the demo of their project and answer questions. Therefore, you should decide ahead which team member will be responsible to demonstrate which part.

You shall start with a brief presentation on the assumptions you have made, E-R Diagram, E-R to relation conversion and normalization. So, prepare a couple of slides accordingly.

Then you will present the functionalities implemented in your project to satisfy the requirements mentioned in the project description. You shall be using the developed application system.

Any additional feature(s) that you have implemented to complement the project can be presented during the demo.